

NATIONAL FORENSIC SCIENCES UNIVERSITY GOA CAMPUS

M.Sc. DFIS - Semester -II Term Assessment-I

Subject Code: CTMSDFIS SII P1

Date: 02/02/2026

Subject Name: Network Security & Forensics

Time: 45 Minutes

Total Marks: 25

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

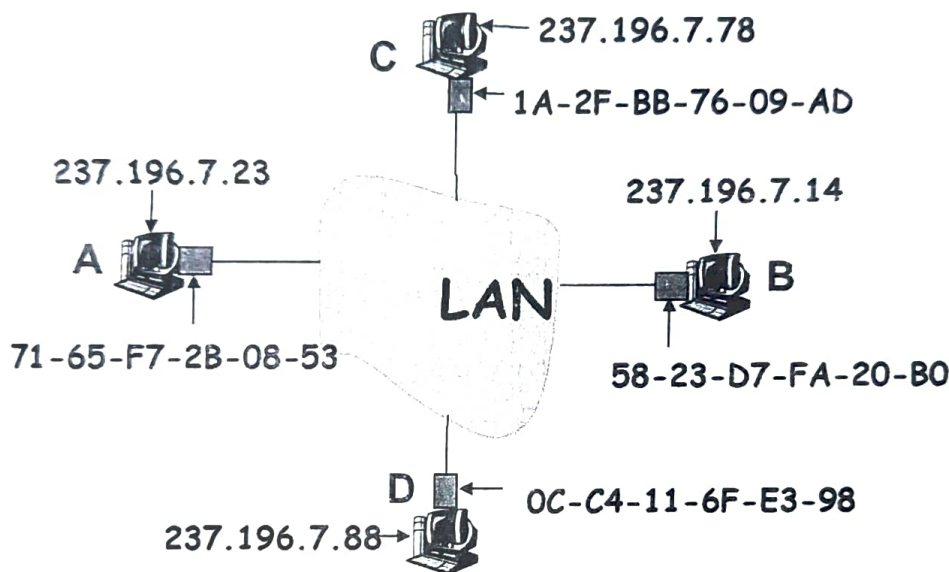
Q01 How does IPv6 differ from IPv4? Highlight the key differences.

05 Marks

Q 02 Describe the components of an Intrusion Detection System (IDS). Analyse the role of Artificial Intelligence in enhancing the efficiency of IDS.

05 Marks

Consider the following Network: (for Q 03-04)



Q 03 Consider the above network topology. User A wants to communicate with User B. Explain the ARP protocol with respect to this scenario.

05 Marks

Q04 Compare TCP and UDP protocols, and describe the purpose of the different TCP flags.

05 Marks

Q 05 Explain the following examples/terms: (Any two)

05 Marks

- (i) Mesh and Star Topology
- (ii) CSMA-CD
- (iii) CSMA-CA